Proposed Solution

Phase 3: Project Design Phase

1. Solution Summary

Trip Genius AI is a web application that converts a traveller’s preferences into an AI-generated itinerary draft. The user interacts with a React and TypeScript frontend. A FastAPI backend validates the request, coordinates the Groq model call and persists the result in Supabase. The interface then provides actions to review, save, reopen, delete and print the plan.

2. User Workflow

1. Open the deployed frontend.
2. Enter starting city, destination, dates, budget, traveller count, interests and travel style.
3. Submit the form and wait for generation to complete.
4. Review the day-wise plan, cost estimates and travel tips.
5. Save the plan if it is useful.
6. Use Saved Trips to reopen or delete a saved itinerary.
7. Open a hotel or restaurant search in Google Maps if more research is needed.
8. Print the plan or select Save as PDF in the browser print dialog.

3. AI Generation

The backend forwards the structured trip inputs to the AI service. The service constructs a prompt that asks for predictable itinerary fields and calls the configured Groq model. The response is parsed as JSON and checked for required keys. This reduces the risk of returning malformed output, although it does not eliminate factual errors or guarantee realistic schedules.

4. Persistence and Saved Trips

The itinerary route saves trip inputs and generated itinerary data to Supabase. Separate endpoints list saved trips, retrieve a selected trip and delete the trip and its associated itinerary. The frontend uses these endpoints to keep the saved-trip workflow integrated with itinerary generation.

5. Hotels and Restaurants Feature

The app builds destination-based Google Maps searches for hotels and restaurants. Restaurant searches can include all, vegetarian, non-vegetarian or Jain preference terms. The backend returns a search URL rather than a list of verified places. Users must inspect the result, confirm opening hours and availability, and verify food preparation directly with the provider.

6. Deployment Model

The frontend is hosted on Vercel and the backend on Render. This split keeps browser presentation separate from API and credential handling. The backend health endpoint provides a simple signal that the API process is responding, but it is not a full test of AI generation or database connectivity.

7. Expected Benefits

- A consistent starting point for travel planning.
- Structured daily activities rather than an unorganized list.
- A saved-trip workflow for later reference.
- A print-friendly way to carry the itinerary.
- Direct transition to external map searches for place research.

8. Limitations

AI-generated plans require human review. Cost values are estimates, and external search links do not verify inventory. Booking, payment, live weather and real-time provider data are not part of the current implementation.